

CHRISTINE E. CRUTE

Duke University, 701 W Main Street, Durham, NC
cef39@duke.edu • (412) 965-5008

EDUCATION

- 2017 – 2022 **Ph.D., Environment, Duke University** (expected May 2022)
Integrated Toxicology and Environmental Health, Nicholas School of the Environment
Division of Reproductive Science, School of Medicine
Thesis: *Maternal exposure to per- and polyfluoroalkyl substances (PFAS) leads to adverse maternal and fetal health outcomes via disruption of placental development and function*
- Graduate Certificates, Duke University**
- Global Health Graduate Certificate, *Duke Global Health Institute*
 - Certificate in College Teaching, *Duke Graduate School*
 - Teaching Writing in the Disciplines Certificate, *Thompson Writing Program*
- 2012 – 2016 **B.S., Environmental Science, George Mason University (GMU) Honors College**
Minor in biology; graduated *summa cum laude*

RESEARCH AND PROFESSIONAL EXPERIENCE

- 2017 – current **Doctoral Student Researcher**, *Division of Reproductive Sciences, Duke University, Durham, NC*
- Implemented novel placental stem cell model to study effects of toxicants on early placentation and fetal health outcomes
 - Optimized and coordinated rabbit model for investigating pregnancy outcomes
- 2019 **Team Lead, China Field and Laboratory Research**, *Duke Global Health Institute and Duke Kunshan University, Shanghai and Taizhou, China*
- Headed laboratory and field research investigating adverse maternal and child health outcomes from exposure to electronic waste recycling processes
- 2016 – 2017 **NIH Post Baccalaureate Research Fellow**, *Laboratory of Malaria and Vector Research, Malaria Genetics Division, National Institute of Allergy and Infectious Disease, Bethesda, MD*
- Investigated drug resistance genetic markers in *Plasmodium falciparum* via genetic crosses utilizing a nonhuman primate model
- 2014 – 2016 **Undergraduate Student Researcher**, *Hamdan Laboratory of Microbial Ecology, George Mason University, Fairfax, VA*
- Designed an Independent Undergraduate Research Project to evaluate the ecological impact of the Deepwater Horizon oil spill and dispersant usage
- 2014 – 2015 **Assistant Field Researcher**, *GOM-SCHEMA Research Cruise, Naval Research Laboratory and George Mason University, Gulf of Mexico*
- Collected and processed water and sediment samples for quantitative analysis

TEACHING AND MENTORING

Teaching Experience

- 2020 – 2021 **Instructor of Record**, *Global Health 395: Impacts of E-waste Exposure on Maternal and Child Health, Duke University Bass Connections, Durham, NC*
- Charted new course focused on e-waste recycling through the toxicological, environmental, community, and political perspectives
 - Formalized syllabus and course materials, orchestrated discussions, and managed student progress
- 2019 **Teaching Assistant**, *Responsible Conduct of Research Orientation Retreat, Duke University, Durham, NC*
- Facilitated discussions about ethical research and best practices in the lab
- 2016 – 2017 **Health Education Instructor**, *Stepping Stones Family Shelter, Rockville, MD*
- Taught general public health lessons to empower medically underserved women to make positive decisions regarding physical and mental health care
- 2013 – 2016 **Peer Instructor**, *University 100, Center for Academic Advising, Resources, and Transitions (CAART), George Mason University, Fairfax VA*
- Co-taught first year course with faculty member
 - Designed syllabus and lesson plans, programmed classroom activities, and mentored first year students
- 2013 – 2015 **Student Lead**, *Project Peak, CAART, George Mason University, Fairfax VA*
- Headed outdoor orientation trip for first year students: facilitated discussions about transition to university life, implemented teambuilding activities

Course Lectures

- 2021 **Elon University**, *BIO 111– Introductory Cell Biology: Gene Expression and Regulation*
- 2020 **Elon University**, *BIO 324 – Toxicology: Per- and polyfluoroalkyl substances (PFAS) contamination in North Carolina drinking water: A case study of current developmental biology toxicity studies.*
- 2019 **Duke University**, *ENV 819 – Mechanisms of Environmental Toxicology: Mechanisms of Toxicity in the Placenta*

Professional Development in Teaching

- 2021 – 2022 **Teaching Writing in the Disciplines (WID) Training**, *Thompson Writing Program*
- Engaged in workshops to train instructors on designing effective writing assignments and providing feedback
 - Building WID portfolio, consisting of reflections on trainings and reading and examples of writing assignments and student feedback
- 2020 – 2021 **Preparing Future Faculty Fellow**, *Duke University*
- Engaged in experiential activities and colloquia to understand faculty roles and responsibilities and prepare for the transition to academia
 - Mentored by Dr. Linda Niedziela, Elon University

- 2020 – 2021 **Courses in College Teaching Pedagogy, Duke University Graduate School**
- GS 760: College Teaching and Visual Communication
 - GS 750: Fundamentals of College Teaching
- 2020 **Teaching Triangles, Duke University Graduate School**
- Provided and received observation during course instruction to reflect and improve on teaching approach and classroom management

Students Mentored

- 2021 Katherine Wang, Duke University Biology, Class of 2023 – Summer research student
- 2021 Henry Che, University of California, Los Angeles Biology, Class of 2023 – Summer research student
- 2019 – 2020 Maya Iskandarani, Duke University Biology, Class of 2020 – Independent study
- 2019 – 2020 Angela Pham, Duke University Biology, Class of 2020 – Independent study
- 2019 Yan Sun, Duke University Masters of Environmental Management, Class of 2020 – China field research
- 2019 Connie Xiong, Duke University Masters of Environmental Management, Class of 2020 – China field research
- 2019 Amy Zhao, Duke University Biology, Class of 2022 – China lab research
- 2019 Emi Yuan, Duke University Biology, Class of 2024 – Summer research student
- 2017 – 2018 Ester Son, North Carolina School of Science and Mathematics, Class of 2018 – High school research program

PROFESSIONAL TRAINING

Science Communication

- 2021 PFAS Communications Workshop & Competition, *Silent Spring Institute*
- 2019 – 2020 IMPACTS Fellow, *UNC's Morehead Planetarium & Science Center*
- 2019 Science Communication Workshop for Graduate Students, *ComSciCon – Triangle 2019*
- 2018 Short Course: Science Communication, *Duke Doctoral Academy*
- 2018 Short Course: Science Policy, *Duke Doctoral Academy*

Professional development

- 2020 Resiliency in Science Training, *Office of Intramural Training and Education, NIH*
- 2020 Short Course: Leading Teams, *Duke Doctoral Academy*
- 2020 Short Course: Business Fundamentals for Entrepreneurs, *Duke Doctoral Academy*
- 2020 Short Course: Public Speaking in a Virtual World, *Duke Doctoral Academy*

Environmental Justice and Health Disparities Training

- 2021 Environmental Justice Symposium, *Duke Undergraduate Environmental Union*
- 2019 Environmental Justice Symposium: Resilient Communities, *Duke Environmental Law and Policy Clinic*
- 2016 – 2017 NIH Academy of Health Disparities Education, *Office of Intramural Training and Education, National Institutes of Health*

RESEARCH COMMUNICATION

- 2020 **Project Team Profile: E-waste Impact on Birth Outcomes**, *Bass Connections, Duke University*. <https://bassconnections.duke.edu/virtual-showcase/e-waste-impact-birth-outcomes>.
- 2019 **Four Lessons from a Novice Field Researcher**. *Voices of DGHI, Duke Global Health Institute*. <https://globalhealth.duke.edu/media/blogs/voices-of-dghi/four-lessons-novice-field-researcher>.
- 2019 **Students in the Field**, *Voices of DGHI, Duke Global Health Institute*. <https://globalhealth.duke.edu/media/blogs/voices-of-dghi/students-field-christine-crute>.

PUBLICATIONS

1. William P. Marinello., Zahra S. Mohseni, Sarah Cunningham, **Christine E. Crute**, Rong Huang, Jun J. Zhang, Liping Feng. (2020). Perfluorobutane sulfonate exposure disrupted human placental cytotrophoblast cell proliferation and invasion involving in dysregulating preeclampsia related genes. *The FASEB Journal*, 00, 1–18. <https://doi.org/10.1096/fj.202000716RR>.
2. Juliana M. Sá, Sarah R. Kaslow, Michael A. Krause, Viviana A. Melendez-Muniz, ... **Christine E. Figan** ... Thomas E. Wellems (2018). Artemisinin resistance phenotypes and K13 inheritance in a *Plasmodium falciparum* cross and *Aotus* model. *Proceedings of the National Academy of Sciences*, 201813386. <https://doi.org/10.1073/pnas.1813386115>.
3. **Christine E. Figan**, Juliana M. Sá, Jianbing Mu, Viviana A. Melendez-Muniz, Chia Hao Liu, Thomas E. Wellems (2018). A set of microsatellite markers to differentiate *Plasmodium falciparum* progeny of four genetic crosses. *Malaria Journal*, 17(1), 60. <https://doi.org/10.1186/s12936-018-2210-z>.

RESEARCH PRESENTATIONS

- 2021 **Christine E. Crute**, Samantha Hall, Chelsea Landon, Angela Garner, Susan K. Murphy, Liping Feng. *Maternal exposure to an environmental per- and polyfluoroalkyl substances (PFAS) mixture leads to adverse pregnancy outcomes in a New Zealand White rabbit model*. Triangle Consortium for Reproductive Biology, poster presentation. February 26, 2021. *Won Best Poster Presentation
- 2020 Ashley Choi, Nadrat Chowdhury, **Christine E. Crute**, Ellea Lamb, Julia Murphy, Angela Pham, Aneesha Raj, Yan Sun, Connie Xiong, Amy Zhao, Liping Feng. *Impacts of E-waste Exposure on Maternal and Fetal Health*. Fortin Foundation Bass Connections Virtual Showcase, poster presentation. April, 2020.
- 2019 **Christine E. Crute**, William P. Marinello, Sarah Cunningham, Zahra Mohseni, Claudia Ambrose, Liping Feng. *Transcriptomic Analysis Provides Molecular Evidence for the Association of Perfluorobutane Sulfonate (PFBS) Exposure and Preeclampsia*. Annual Meeting of the U.S. Developmental Origin of Health and Disease (DOHAD) Society, poster presentation. Chapel Hill, NC. September 9, 2019.
- 2017 **Christine E. Figan**. *A set of microsatellite markers to differentiate Plasmodium falciparum progeny from four genetic crosses*. Laboratory of Malaria and Vector

Research Seminar Series, oral presentation. National Institutes of Allergy and Infectious Disease, July 14, 2017.

2017 **Christine E. Figan**, Juliana M. Sá, Jianbing Mu, Viviana A. Melendez-Muniz, Chia Hau Liu, Thomas E. Wellems. *Fingerprinting Malaria Parasites through Microsatellite Genotyping*. National Institutes of Health Postbac Poster Day, poster presentation. Bethesda, MD. May 4, 2017.

ACADEMIC FELLOWSHIPS, AWARDS, FEATURES

2021	Campion Fund Award, Best Poster Presentation by a Junior Investigator (\$2,500)
2021	Charles Hammond Research Fund Award, Duke University Medical Center (\$10,000)
2021	US DOHAD Society Newsletter, Featured Trainee Scientist
2019	DOHAD Trainee Travel Award (conference fee)
2019	Fieldwork Travel Award, Duke Global Health Institute (\$5000)
2017 – 2019	NIH T32, Duke University Program in Environmental Health (tuition & stipend)
2016	Outstanding Environmental Health Undergraduate Award, GMU (\$1000)
2013 – 2015	Provost Scholarship, GMU (tuition support)

SERVICE AND OUTREACH

2021 – present	Leadership Committee, PhD Representative , <i>Environmental Communication, Outreach, and Education (ECOE) Club, Duke University</i>
2019 – 2021	Organizing Committee , <i>ComSciCon – Triangle 2020 Workshop</i>
2020	Healthcare COVID19 Care Package Coordinator , <i>OBGYN, Duke University</i>
2018 – 2019	Graduate School Representative, Community Service Committee , <i>Graduate and Professional Student Council, Duke University</i>
2018	Event Lead, Science Communication for STEM Graduate Students , <i>Duke Women in Science and Engineering (WiSE)</i>
2015 – 2016	Project Manager & Field Volunteer (Iquitos, Peru) , <i>Amazon Water, Sanitation, and Hygiene (WaSH) Student Organization, GMU</i>